

KAMPIBIOTIC 500 Injection

Ampicillin 500 mg

Product Name

KAMPIBIOTIC 500 Injection

Product Description

White powder in glass vials.

Composition

Each vial contains Ampicillin sodium BP equivalent to Ampicillin 500 mg

Pharmacodynamics

Ampicillin exerts its action on growing and dividing bacteria by inhibiting bacterial cell wall synthesis. Benzylpenicillin inhibits the final cross-linking stage of peptidoglycan production by binding to and inactivating transpeptidases (penicillin-binding protein on the surface of the bacterial cell membrane) causing the cell wall to rupture by osmotic pressure. It is now realised that other earlier stages in cell-wall synthesis can also be inhibited by Ampicillin.

Pharmacological action

Ampicillin is bactericidal for both Gram-positive and Gram-negative organisms.

Gram-Positive Organisms: Ampicillin is active against *Strep. Pyogenes*, *Strep. Pneumoniae* and *Strep. Viridans*. Pneumococcal strains which are relatively resistant to penicillin G are more sensitive to Ampicillin. Group B Streptococci are always Ampicillin sensitive, in addition Ampicillin combines with aminoglycosides such as Kanamycin, gentamicin, tobramycin or amikacin acts synergistically against these organisms resulting in accelerated streptococcal killing.

Strep. faecalis is usually sensitive to Ampicillin and often more so than penicillin G. The combination of ampicillin with streptomycin or kanamycin is synergistic against 50-75% of *Strep. faecalis*, whereas Ampicillin plus gentamicin or tobramycin is synergistic against nearly all such strains. Anaerobic Gram-positive cocci such as peptococcus and peptostreptococcus spp. and anaerobic streptococci are nearly always Ampicillin sensitive.

Ampicillin is destroyed by staphylococcal beta lactamase, so that most *Staph. aureus* and hospital acquired *Staph. epidermidis* are Ampicillin resistant. *Staph. saprophyticus* and other coagulase negative staphylococci such as *Staph. hominis* and haemolyticus which may be urethral isolates of young females are Ampicillin sensitive. *C. diphtheriae* and *B. anthracis* are sensitive to Ampicillin. *L. monocytogens* is also usually sensitive to Ampicillin. Most Nocardia spp. strains are Ampicillin resistant. Anaerobic Gram-positive sporing bacillus such as *Cl. Tetani*, *Cl. Perfringens* (welchii), *Cl. Botulinum* and other Clostridium spp. are usually Ampicillin sensitive.

Ampicillin is nearly always active against anaerobic Gram-positive asporogenous bacilli such as Actinomyces, Eubacterium, Arachnia, Propionibacterium, Bifidobacterium and Lactobacillus.

Gram-negative Organism

Ampicillin is active against many enterobacteriaceae. *E. coli* may be sensitive but many strains are resistant. Ampicillin resistance of *E.coli* is nearly always due to beta lactamase production. *Proteus mirabilis* is usually sensitive unless it is a beta lactamase producing strain.

Salmonella spp. is usually susceptible to Ampicillin. The Brucella spp. and *B. pertussis* are consistently sensitive to Ampicillin. The same is true of *No. meningitides*. *Branhamella catarrhalis* is Ampicillin sensitive except strains producing beta lactamase. *V. cholerae* and *Campylobacter jejuni* are usually sensitive. *Gardenerella vaginalis* is Ampicillin sensitive.

Some Gram-negative anaerobic bacteria such as *B. melaninogenicus* and Fusobacterium spp. are usually Ampicillin sensitive. Other bacteroides spp. vary in sensitivity but about 50% of isolates are inhibited by low concentrations.

Pharmacokinetics

Absorption and Serum level

Ampicillin is given by injection as the sodium salt. After intramuscular administration of 500 mg, a serum peak of 7-14 mg /lit can be expected after 1 hour.

The half life of Ampicillin is 1-2 hours and the plasma protein binding is around 20%.

Distribution

Ampicillin is distributed throughout the body and is to be found at lower concentration than in serum at most sites with the exception of the liver and kidneys. Penetration of the blood-barrier is poor except during active inflammation when a daily dose of 150 mg Ampicillin per kg may result in CSF level around 3 mg/lit.

Adequate therapeutic levels are reached in pleural, synovial and ocular fluids. Ampicillin crosses amniotic fluids during the first trimester do not reach therapeutic concentration. Some Ampicillin is secreted in the milk of a nursing mother.

Metabolism

Hepatic metabolism is a relatively unimportant route of Ampicillin elimination. About 10% of the dose is metabolised in the liver to the inactive penicilloic acid. Ampicillin esters are metabolised in the gut to the active form of the drug.

Excretion

Renal clearance of Ampicillin is slower than that of benzylpenicillin, accounting for the longer half life. Renal excretion is both by glomerular filtration & by tubular secretion. At normal oral doses 30% Ampicillin is excreted in the urine resulting in levels between 250 mg/l & 75% of the dose may be recovered unchanged from the urine in the same period. This can be reduced by simultaneous dosage with probenecid. The half life of

Ampicillin may be prolonged in patients with renal failure through no adjustment of dose is likely to be necessary, other than in very severe cases. Ampicillin is also excreted in the bile, but levels are variable & have a particular tendency to be low in the presence of a non-functional gallbladder. There is a small enterohepatic circulation of the drug & the unexcreted agent is inactivated in the liver at a slower rate than Benzylpenicillin. The kinetics of Ampicillin are unlikely to be altered in patients with hepatic dysfunction.

Plasma half life range	1-2 hour
Volume of distribution	0.38 lit/kg
Plasma protein binding	20%

Acute Toxicity

Long term studies have not revealed any serious toxicity associated with Ampicillin. Toxicology and carcinogenesis studies of Ampicillin were performed in F344/N rats and B6C3F1 mice. In these studies Ampicillin was administered for 2 years to rats at doses of 0, 750 or 1500 mg and to mice at doses 0, 1500 or 3000 mg/kg. The drug was administered by oral gavage in corn oil. No toxic manifestations other than toxic lesions of the stomach were seen in rats and mice after Ampicillin administration.

Indication

Ampicillin is a broad spectrum penicillin, indicated for the treatment of a wide range of bacterial infections caused by ampicillin sensitive organisms. Typical indications include: ear, nose and throat infections, bronchitis, pneumonia, urinary tract infections, gonorrhoea, gynaecological infections, septicaemia, peritonitis, endocarditis, meningitis, enteric fever, gastro-intestinal infections.

Extraperitoneal application of Ampicillin to wounds can be used to prevent infection following abdominal surgery.

Parenteral usage is indicated where oral dosage is inappropriate.

Recommended Dosage

Usual adult dosage (including elderly patients):

Septicaemia, endocarditis, osteomyelitis: 500 mg four to six times a day IM or IV for one to six weeks.

Peritonitis, intra-abdominal sepsis: 500 mg four times a day IM or IV. Meningitis:

Adult dosage: 2 g six-hourly IV.

Children's dosage: 150 mg / kg daily IV in divided doses. Ampicillin may also be administered by other routes of conjunction with systemic therapy.

Intraperitoneal: 500 mg daily in up to 10 ml Water for Injections BP.

Intrapleural: 500 mg daily in 5-10 ml water for Injections BP. Intra-articular: 500 mg daily, in up to 5 ml Water for Injections BP or sterile 0.5% procaine hydrochloride solution.

Local use in abdominal surgery: 1 g sterile powder sprinkled into the wound extraperitoneally or into muscle layers to prevent wound infection post-operatively.

Usual children's dosage (under 10 years):

Half adult routine dosage.

All recommended dosages are a guide only. In severe infections the above dosages may be increased.

Renal Impairment:

In the presence of severe renal impairment (creatinine clearance $<10\text{ml/min}$) a reduction in dose or extension of dose interval should be considered. In cases of dialysis, an additional dose should be administered after the procedure.

Administration:

Intramuscular: Add 1.5 ml Water for Injections BP to 500mg vial contents. Intravenous: Dissolve 500 mg in 10 ml Water for Injections BP. Administer by slow injection (three to four minutes). Ampicillin may also be added to infusion fluids or injected, suitably diluted, into the drip tube over a period of three to four minutes.

Incompatibility

Extemporaneous admixtures of beta-lactam antibacterials (penicillins and cephalosporins) and aminoglycosides may result in substantial mutual inactivation. If these groups of antibacterials are administered concurrently, they should be administered in separate sites at least 1 hour apart. Do not mix them in the same intravenous bag, bottle or tubing.

When aminoglycosides and penicillins are administered separately by different routes, a reduction of aminoglycoside serum concentration may be impaired renal function when excretion of both medications is delayed.

Contraindications

Serious and occasionally fatal hypersensitivity (anaphylactic) reactions have been reported in patients on penicillin therapy. Anaphylaxis is more frequent following parenteral therapy than oral penicillins. These reactions are more apt to occur in individuals with a history of sensitivity to multiple allergens.

Penicillin Hypersensitivity

There is cross-allergenicity between ampicillin and other penicillins, so that patients with a history of penicillin allergy may respond similarly to Ampicillin. Cross-allergenicity also occurs between penicillins and cephalosporins in a proportion of patients.

Warnings and Precautions

Before initiating therapy with ampicillin, careful enquiry should be made concerning previous hypersensitivity reactions to beta-lactam antibiotics.

Serious and occasionally fatal hypersensitivity reactions (anaphylaxis) have been reported in patients receiving beta-lactam antibiotics. Although anaphylaxis is more frequent following parenteral therapy, it has occurred in patients on oral penicillins.

These reactions are more likely to occur in individuals with a history of beta-lactam hypersensitivity.

Ampicillin should be avoided if infectious mononucleosis and / or acute or chronic leukaemia of lymphoid origin are suspected. The occurrence of a skin rash has been associated with these conditions following the administration of ampicillin.

Prolonged use may occasionally result in overgrowth of non-susceptible organisms. Dosage should be adjusted in patients with renal impairment.

Interactions with Other Medicaments

If Ampicillin is prescribed concurrently with an aminoglycoside, the antibiotics should not be mixed in the syringe, intravenous fluid container or giving set because loss of activity of the aminoglycoside can occur under these conditions.

Bacteriostatic drugs may interfere with the bactericidal action of ampicillin. In common with other oral broad-spectrum antibiotics, ampicillin may reduce the efficacy of oral contraceptives and patients should be warned accordingly.

Probenecid decreases the renal tubular secretion of ampicillin. Concurrent use with ampicillin may result in increased and prolonged blood levels of ampicillin. Concurrent administration of allopurinol during treatment with ampicillin can increase the likelihood of allergic skin reactions.

It is recommended that when testing for the presence of glucose in urine during ampicillin treatment, enzymatic glucose oxidase methods should be used. Due to the high urinary concentrations of ampicillin, false positive readings are common with chemical methods.

Pregnancy and Lactation

Pregnancy

Animal studies with Ampicillin have shown no teratogenic effects. When antibiotic therapy is required during pregnancy, Ampicillin may be considered appropriate.

Lactation:

During lactation, trace quantities of penicillins can be detected in breast milk. Adequate human and animal data on use of Ampicillin during lactation are not available.

Side Effects

Hypersensitivity reactions:

If any hypersensitivity reaction occurs, the treatment should be discontinued. Skin rash, pruritis and urticaria have been reported occasionally. The incidence is higher in patients suffering from infectious mononucleosis and acute or chronic leukaemia of lymphoid origin. Purpura has also been reported. Rarely, skin reactions such as erythema multiforme and Stevens-Johnson syndrome, and toxic epidermal necrolysis have been reported.

As with other antibiotics, anaphylaxis has been reported rarely.

Renal effects:

Interstitial nephritis can occur rarely.

Gastrointestinal reactions:

Effects include nausea, vomiting and diarrhoea. Pseudomembranous colitis and haemorrhagic colitis has been reported rarely.

Hepatic effects:

As with other beta-lactam antibiotics, hepatitis and cholelstatic jaundice have been reported rarely. As with most other antibiotics, a moderate and transient increase in transaminases has been reported.

Haematological effects:

As with other beta-lactams, haematological effects including transient leucopenia, transient thrombocytopenia and haemolytic anaemia have been reported rarely. Prolongation of bleeding time and prothrombin have also been reported rarely.

Symptoms and Treatment of Overdose

Gastrointestinal effects such as nausea, vomiting and diarrhoea may be evident and should be treated symptomatically.

Ampicillin may be removed from the circulation by haemodialysis

Presentation

One folded paper box contains 10 vials or 50 vials.

Storage Condition

Store below 30°C. Protect from light. Keep out of reach of children.

Shelf Life

36 months from the date of manufacture if kept as recommended.

Registration number

KAMPIBIOTIC 500 Injection - MAL07061439AZ

Name and Address of Manufacturer

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